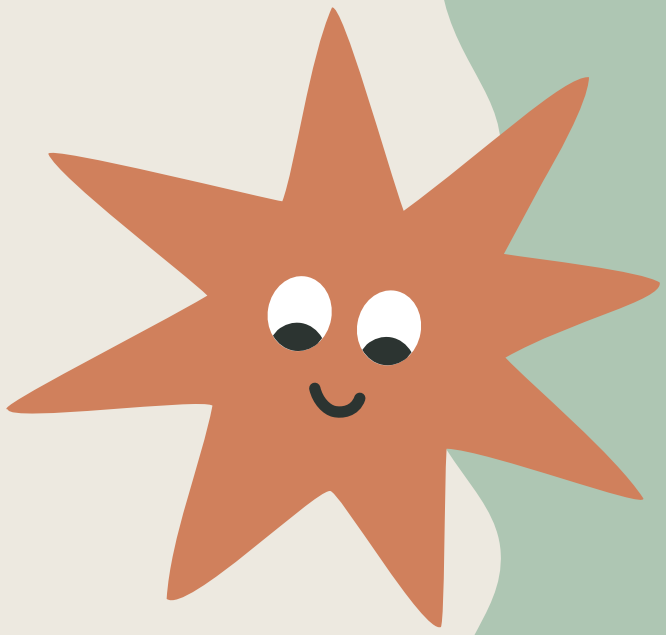
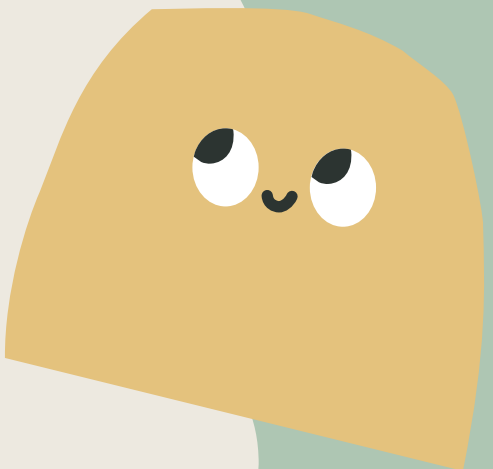
Loops in Python

Lecture 6





Learning Objectives

➢By the end of this lecture, students should be able to:

- Understand why loops are needed in programming
- Use for loops to repeat tasks
- Use range() to generate sequences of numbers
- Use while loops when repetition depends on a condition
- Use break to stop a loop
- Use continue to skip the current iteration
- Understand the difference between for and while
- Apply loops to practical programming and simple AI-related problems

Prepared By: Ehtisham Rasheed

What is a Loop?

- A **loop** is a programming structure that repeatedly executes a block of code.
- Loops are extremely important in AI and data science because we frequently need to process:
  - Many students
  - Many numbers
  - Many images
  - Many text documents
- For example, if an AI model has 10,000 training examples, we need some mechanism to process them repeatedly.

Prepared By: Ehtisham Rasheed

Types of Loops in Python

➢Python primarily provides two types of loops:

1. **for** loop
   - Used when we want to iterate over a sequence or collection

```
for item in sequence:
    # code
```

2. **while** loop
   - Used when we want to repeat something **while a condition remains true**

```
while condition:
    # code
```

Prepared By: Ehusham Rasheed

The for Loop

- The for loop is one of the most commonly used loops in Python.
- Basic syntax

```
for variable in sequence:
    statement
```

- Example:

```
for i in [1, 2, 3, 4, 5]:
    print(i)
```

```
name = "Python"

for character in name:
    print(character)
```

Prepared By: Eng. Ibtisham Rasheed

Using range()

- One of the most important functions used with for loops is:
  - range()
  - It generates a sequence of numbers.
- Example:

```
for i in range(5):
    print(i)
```

- range(start, stop)

```
for i in range(1, 6):
    print(i)
```

range(start, stop, step)

```
for i in range(1, 11, 2):
    print(i)
```

Counting Backwards

```
for i in range(10, 0, -1):
    print(i)
```

Presented By: Ehtisham Rasheed

Calculating a Sum Using a Loop

- Suppose we want to calculate:

$$1 + 2 + 3 + 4 + 5$$

- We can write:

```
total = 0

for i in range(1, 6):
    total = total + i

print("Total:", total)
```

Prepared by Ehtisham Rasheed

Looping Through a List

- Although lists will be covered more formally later, you can understand this simple example.

```
students = ["Ali", "Ahmed", "Sara", "Ayesha"]

for student in students:
    print(student)
```

- We can use conditional statements inside loop as well:

```
for number in range(1, 11):

    if number % 2 == 0:
        print(number, "is even")
```

Prepared By: E. Rasheed

The while Loop

- The second major loop in Python is:
  - while
- A while loop repeats code **as long as a condition is true**.
- Syntax:

```
while condition:
    statement
```

- Example:

```
count = 1

while count <= 5:
    print(count)
    count = count + 1
```

Prep... BY: Ehtisham Rasheed

while Loop — User Input

- A while loop is useful when we don't know exactly how many times something should happen.

```
password = ""

while password != "python123":
    password = input("Enter password: ")

print("Access granted!")
```

Prepared by Mohammed Ihtisham Rasheed

The break Statement

- Sometimes we want to **stop a loop immediately**.
- For this, Python provides **break** statement.
- Example:

```
for i in range(1, 11):

    if i == 5:
        break

    print(i)
```

```
numbers = [10, 20, 30, 40, 50]

for number in numbers:

    if number == 30:
        print("Number found!")
        break
```

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The continue Statement

- **continue** is different from break.
- **continue** skips the current iteration and moves to the next iteration.
- Example:

```
for i in range(1, 6):

    if i == 3:
        continue

    print(i)
```

```
marks = [78, -5, 90, 65, -10, 88]

for mark in marks:

    if mark < 0:
        continue

    print("Valid mark:", mark)
```

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AI/Data Processing Example

➢Suppose an AI system receives data:

```
data = [10, 20, None, 30, None, 40]
```

➢We don't want to process missing values.

```
for value in data:

    if value is None:
        continue

    print("Processing:", value)
```

Prepared by Shubham Rasheed

Combining for, if, break, and continue

➢We can combine everything

```
numbers = [10, -5, 20, 30, -2, 40, 50]

for number in numbers:

    if number < 0:
        continue

    if number == 40:
        break

    print(number)
```

Output

```
10
20
30
```

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Nested Loops

- A loop can exist inside another loop.
- Example:

```
for i in range(3):

    for j in range(3):
        print(i, j)
```

- This is called a **nested loop**.
- Nested loops become very important when working with:
  - Tables
  - Matrices
  - Grids
  - Images

Prepared By: Ehti

Intermediate Practice

- Exercise 1 — Positive/Negative

```
numbers = [10, -5, 20, -2, 30, 40]
```

- Exercise 2 — Skip Zero

```
numbers = [10, 0, 20, 0, 30]
```

- Exercise 3 — Stop at Negative

```
numbers = [10, 20, 30, -5, 40, 50]
```

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Challenge 1 — Student Statistics

➢Given:

```
marks = [78, 45, 89, 32, 91, 67, 55, 40]
```

➢Calculate:

- Total students
- Number of passed students
- Number of failed students
- Highest marks
- Lowest marks
- Average marks

Prepared By: Ehtisham Rasheed

Challenge 2 — Simple AI Alert System

➢Given:

```
temperatures = [25, 28, 32, 35, 38, 42, 45]
```

➢Create a system that:

- Prints Normal below 30
- Prints Warning from 30–39
- Prints Critical at 40+
- Stops processing when the first critical value appears

Prepared By: Ehtisham Rasheed